University of Ottawa

Department of Mathematics and Statistics

MAT 1302C : Mathematical Methods II

Professor: Hadi Salmasian

Final Exam

December 11, 2011

Surname _____ First Name _____

Student # _____ DGD (Tuesday/Thursday) _____

Instructions:

(a) You have three hours to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1–14	15	16	17	18	19	20	21	22	Total
Maximum	28	4	4	5	6	7	4	6	4	68
Grade										

Part A: Answer Only Questions

In Questions 1–14, only your final answer will be considered for marks. Each question is worth 2 points.

1. Let $A = \begin{bmatrix} 2 & -1 & 3 \\ -2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 1 & 1 \end{bmatrix}$. Calculate AB.

Solution: $\begin{bmatrix} 2 & -1 & 3 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & -3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ -3 & 1 \end{bmatrix}$

2. Suppose that $A = \begin{bmatrix} 3 & 0 & -1 & 3 \\ 0 & -1 & 12 & -4 \\ 0 & 0 & 2 & -\frac{8}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & \frac{1}{4} & 0 & 0 \\ -\frac{2}{3} & 1 & \frac{1}{4} & 0 \\ -1 & 2 & 1 & -1 \end{bmatrix}$. Calculate $\det(2AB)$.

Solution: $\det(2AB) = 2^4 \det A \det B = 16(-6)(-\frac{3}{16}) = 18$

3. Let A be an $m \times n$ matrix. Which of the following statements implies that the matrix equation $A\mathbf{x} = \mathbf{b}$ has a solution for every vector $\mathbf{b}$ in $\mathbb{R}^m$?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $\operatorname{Col} A = \mathbb{R}^m$.
(b) A has at least n pivot positions.
(c) $m < n$.
(d) Every row of A contains a pivot position.
(e) $\operatorname{Nul} A = \mathbb{R}^n$.

Solution: (a), (d).

4. Let A be a 8×17 matrix. We know that $\dim \operatorname{Col} A = 7$. Which of the following statements is definitely false?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) For every column vector $\mathbf{b}$ in $\mathbb{R}^8$, the linear system $A\mathbf{x} = \mathbf{b}$ has a solution.
(b) The columns of A are linearly dependent.
(c) $\dim \operatorname{Nul} A = 1$.
(d) The homogeneous system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
(e) The pivot columns of A are linearly independent.

Solution: (a), (c)

5. Let $A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 1 & -\frac{1}{3} & -2 \end{bmatrix}$. Write down the eigenvalues of A and their multiplicities.

Solution: -2 with multiplicity 2,
3 with multiplicity 1.

6. Compute the inverse of the matrix $\begin{bmatrix} 1+i & -1 \\ -1 & 1 \end{bmatrix}$. The entries of your answer should be complex numbers of the form $a + bi$ where a, b are real numbers.

Solution: $\begin{vmatrix} 1+i & -1 \\ -1 & 1 \end{vmatrix} = i \neq 0$ and therefore the matrix is invertible with inverse $\frac{1}{i}\begin{bmatrix} 1 & 1 \\ 1 & 1+i \end{bmatrix} = \begin{bmatrix} -i & -i \\ -i & 1-i \end{bmatrix}$.

7. Suppose A, B and C are 3×3 matrices such that $\det A = -2$, $\det B = -\frac{1}{3}$ and $\det C = -1$. Calculate $\det\left((-3)A^{-1}(B^2)^T C A^{-2}\right)$.

Solution: $\det\left((-3)A^{-1}(B^2)^T C A^{-2}\right) = (-3)^3(-\frac{1}{2})(-\frac{1}{3})^2(-1)(-\frac{1}{2})^2 = -\frac{3}{8}$.

8. Which of the following sets is a subspace of $\mathbb{R}^n$ for the given value of n?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $H = \left\{ \begin{bmatrix} x-2y \\ y+z \\ z-t \end{bmatrix} \;\middle|\; x, y, z, t \text{ are in } \mathbb{R} \right\}$ and $n = 3$.

(b) $H = \operatorname{Nul} A$ where $A = \begin{bmatrix} 1 & 3 & -2 & 6 & 1 \\ 0 & -3 & 3 & 1 & 5 \\ 2 & 2 & 1 & 0 & 1 \end{bmatrix}$ and $n = 3$.

(c) $H = \left\{ \begin{bmatrix} x \\ y^2 z \\ 0 \end{bmatrix} \;\middle|\; x, y, z \text{ are in } \mathbb{R} \right\}$ and $n = 3$.

(d) $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$ and $n = 3$.

(e) $H = \left\{ \begin{bmatrix} a \\ 0 \\ -1 \\ b \end{bmatrix} \;\middle|\; a, b \text{ are in } \mathbb{R} \right\}$ and $n = 4$.

Solution: (a), (c), (d).

9. Suppose a, b, c are numbers such that:

$$\begin{vmatrix} a & 0 & 3 & -2 \\ 1 & b & -2 & -4 \\ 2 & -5 & 0 & c \\ 1 & 1 & -2 & -1 \end{vmatrix} = -3.$$

Calculate $\begin{vmatrix} a+2 & -5 & 3 & c-2 \\ \frac{1}{2} & \frac{b}{2} & -1 & -2 \\ 2 & -5 & 0 & c \\ -2 & -2 & 4 & 2 \end{vmatrix}$.

Solution: The row operations to get from the first matrix to the second are: $R_1 \to R_1 + R_3$, $R_2 \to \frac{1}{2}R_2$ and $R_4 \to (-2)R_4$. Therefore the answer is 3.

10. Is the vector $\begin{bmatrix} -1 \\ 1+i \end{bmatrix}$ an eigenvector of the matrix $\begin{bmatrix} -i+1 & 1 \\ -4 & -2 \end{bmatrix}$? If the answer is yes, determine the associated eigenvalue.

Solution: $\begin{bmatrix} -i+1 & 1 \\ -4 & -2 \end{bmatrix}\begin{bmatrix} -1 \\ 1+i \end{bmatrix} = \begin{bmatrix} 2i \\ 2-2i \end{bmatrix} = (-2i)\begin{bmatrix} -1 \\ 1+i \end{bmatrix}$. Therefore the answer is yes and the eigenvalue is $-2i$.

11. Suppose $A = \begin{bmatrix} -2 & 4 & 5 & -2 & 3 & 6 & -5 \\ 0 & 8 & 0 & -3 & 0 & -1 & -2 \\ 0 & 0 & 0 & -2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. What is the dimension of $\operatorname{Nul} A$?

Solution: The matrix is in E.F. and $\operatorname{rank} A = 3$, therefore by the Rank Theorem we have $\dim \operatorname{Nul} A = 7 - 3 = 4$.

12. Suppose A and B are $n \times n$ matrices. As usual, I_n is the $n \times n$ identity matrix and 0_n is the $n \times n$ zero matrix. Which of the following statements are always true?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) If $AB = \mathrm{I}_n$ then $BA = \mathrm{I}_n$.
(b) If $AB = 0_n$, then $BA = 0_n$.
(c) $\det(AB) = \det(BA)$.
(d) If $\operatorname{rank} AB = n$ then $\operatorname{rank} A = n$.
(e) If $n = 8$ then $\det(-AB) = \det(BA)$.
(f) $\det(A + B) = \det(A) + \det(B)$.

Solution: (a), (c), (d), (e).

13. Calculate the determinant of $\begin{bmatrix} 1 & -3 & 4 & -\frac{1}{4} & 0 \\ 2 & -1 & 4 & -5 & 0 \\ 2 & -1 & 4 & -5 & 0 \\ 0 & 1 & -1 & 0 & -4 \\ -7 & 1 & 5 & 4 & 6 \end{bmatrix}$.

Solution: The answer is 0 because the matrix has two equal rows.

14. Suppose that $z = 1 - i$ and $w = 1 + 2i$. Compute $\frac{w}{\overline{z}}$ and write your answer as a complex number of the form $a + bi$ where a and b are real numbers.

Solution: $\frac{w}{\overline{z}} = \frac{3}{2} + \frac{1}{2}i.$

Part B: Long Answer Questions

In Questions 15–22, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

15. **(4 points)** Is the following linear system consistent or inconsistent? If it is consistent then find its general solution and write it in vector parametric form.

$$\begin{cases} 2x_1 - x_2 - 5 &= x_3 + 4x_4 \\ 2x_2 + x_3 + 6x_4 + 8 &= 2x_1 \\ 4x_1 - 2x_5 - 8 &= 2x_2 + 2x_3 + 8x_4 \end{cases}$$

Solution: We find the R.E.F. of the augmented matrix:

$$\left[\begin{array}{ccccc|c} 2 & -1 & -1 & -4 & 0 & 5 \\ -2 & 2 & 1 & 6 & 0 & -8 \\ 4 & -2 & -2 & -8 & -2 & 8 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_2]{R_2 \to R_2 + R_1} \left[\begin{array}{ccccc|c} 2 & -1 & -1 & -4 & 0 & 5 \\ 0 & 1 & 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 0 & -2 & -2 \end{array}\right]$$

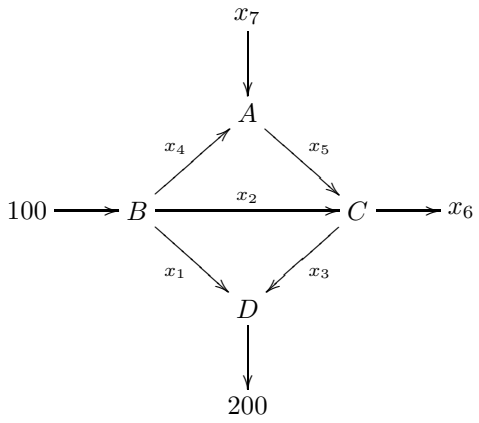
$$\xrightarrow{R_3 \to -\frac{1}{2}R_3} \left[\begin{array}{ccccc|c} 2 & -1 & -1 & -4 & 0 & 5 \\ 0 & 1 & 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 + R_2} \left[\begin{array}{ccccc|c} 2 & 0 & -1 & -2 & 0 & 2 \\ 0 & 1 & 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to \frac{1}{2}R_1} \left[\begin{array}{ccccc|c} 1 & 0 & -\frac{1}{2} & -1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & -3 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array}\right] \quad \Rightarrow \quad \begin{cases} x_1 = \frac{1}{2}x_3 + x_4 + 1 \\ x_2 = -2x_4 - 3 \\ x_3 : \text{free} \\ x_4 : \text{free} \\ x_5 = 1 \end{cases}$$

and the vector parametric form of the solution is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}x_3 + x_4 + 1 \\ -2x_4 - 3 \\ x_3 \\ x_4 \\ 1 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -3 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

16. The traffic flow in a city is represented by the diagram below. The arrows indicate the directions of one-way traffic. Numbers and variables indicate the number of cars per hour in the corresponding direction.



(a) **(3 points)** Write down a linear system describing the traffic flow. Do not solve the linear system at this stage.

Solution:

Total: $x_7 + 100 = x_6 + 200$

A : $x_4 + x_7 = x_5$

B : $100 = x_1 + x_2 + x_4$

C : $x_2 + x_5 = x_3 + x_6$

D : $x_1 + x_3 = 200$

(b) **(1 point)** The reduced echelon form of the augmented matrix of the above linear system is the following matrix:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 200 \\ 0 & 1 & -1 & 0 & 1 & 0 & -1 & -100 \\ 0 & 0 & 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & -100 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Suppose that due to road closures, the maximum number of cars entering the city at A from the north (i.e., along the downward arrow ending at A) is limited to 150 cars per hour. What is the maximum number of cars per hour that leave the city at C towards east (i.e., along the arrow starting at C and pointing to the right)? Justify your answer.

Solution: From the above matrix we have $x_6 = x_7 - 100$. If $x_7 \leq 150$ then $x_6 \leq 50$.

17. (a) **(3 points)** Find all values of t such that the vector $\begin{bmatrix} 5 \\ -6 \\ 9 \end{bmatrix}$ is in Span $\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ t \end{bmatrix} \right\}$.

Solution: We should row reduce the following matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ -1 & -1 & -3 & -6 \\ 2 & 4 & t & 9 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & t-6 & -1 \end{array}\right]$$

The last matrix is in E.F. and to be consistent, the rightmost column should not be a pivot column. It follows that $t \neq 6$.

(b) **(2 points)** Suppose $t = 7$. Find scalars c_1, c_2, c_3 such that

$$\begin{bmatrix} 5 \\ -6 \\ 9 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + c_3 \begin{bmatrix} 3 \\ -3 \\ t \end{bmatrix}.$$

Solution: For $t = 7$ we continue the row reduction:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 3R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 10 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

from which it follows that $c_1 = 10$, $c_2 = -1$ and $c_3 = -1$.

18. The town of Richville has two sports facilities: Arnie's and BestFit. Every physically active resident of Richville is a member of one of the two sports facilities. On January 1, 2011, the market share of Arnie's (i.e., the percentage of Richville's physically active residents which were members of Arnie's) was 40% while the market share of BestFit was 60%. Each month, Arnie's loses $\frac{1}{6}$ of its members to BestFit, while $\frac{1}{3}$ of the members of BestFit switch to Arnie's.

(a) **(1 point)** Write the migration matrix M.

Solution: $M = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{2}{3} \end{bmatrix}$

(b) **(2 points)** On January 1, 2011, the total number of physically active people in Richville was 6000 individuals. How many members did each of the sports facilities have on February 1, 2011?

Solution: $\mathbf{x}_1 = M\mathbf{x}_0 = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} .4 \\ .6 \end{bmatrix} = \begin{bmatrix} \frac{32}{60} \\ \frac{28}{60} \end{bmatrix}$ and therefore Arnie's had $6000 \times \frac{32}{6} = 3200$ members and BestFit had $6000 \times \frac{28}{60} = 2800$ members.

(c) **(3 points)** Assuming that the migration matrix remains constant, compute the long term market share percentages of both of the sports facilities.

Solution: The long term market share is given by the steady state distribution. We have

$$M\mathbf{x} = \mathbf{x} \;\Rightarrow\; (M - \mathrm{I}_2)\mathbf{x} = \mathbf{0} \;\Rightarrow\; \begin{bmatrix} -\frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -\frac{1}{6} & \frac{1}{3} & 0 \\ \frac{1}{6} & -\frac{1}{3} & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} -\frac{1}{6} & \frac{1}{3} & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -\frac{1}{6}R_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right] \;\Rightarrow\; \mathbf{x} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

and the steady state distribution is $\mathbf{p} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$. Therefore the market share of Arnie's is $\frac{2}{3} \approx 66.66\%$ and the market share of BestFit is $\frac{1}{3} = 33.33\%$.

19. Let $A = \begin{bmatrix} 4 & 0 & 7 \\ 1 & -3 & 1 \\ -1 & 0 & -4 \end{bmatrix}$.

(a) **(2 points)** Write the characteristic polynomial of A and compute its eigenvalues.

Solution:

$$\det(A-\lambda I) = \begin{vmatrix} 4-\lambda & 0 & 7 \\ 1 & -3-\lambda & 1 \\ -1 & 0 & -4-\lambda \end{vmatrix} = (-3-\lambda)\begin{vmatrix} 4-\lambda & 7 \\ -1 & -4-\lambda \end{vmatrix}$$
$$= (-3-\lambda)\big((4-\lambda)(-4-\lambda)+7\big) = (-3-\lambda)(\lambda^2-9) = -(\lambda-3)(\lambda+3)^2.$$

Therefore the eigenvalues of A are 3 and -3.

(b) **(4 points)** For each of the eigenspaces of A, find a basis.

Solution:

$$\lambda = 3 \;\Rightarrow\; A-\lambda I = \begin{bmatrix} 1 & 0 & 7 \\ 1 & -6 & 1 \\ -1 & 0 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 7 \\ 1 & -6 & 1 \\ -1 & 0 & -7 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 0 & 7 \\ 0 & -6 & -6 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \to -\frac{1}{6}R_2} \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -7x_3 \\ -x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -7 \\ -1 \\ 1 \end{bmatrix} \Rightarrow \left\{ \begin{bmatrix} -7 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\lambda = -3 \;\Rightarrow A-\lambda I = \begin{bmatrix} 7 & 0 & 7 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 0 & 7 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{7}R_1} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ -1 & 0 & -1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(c) **(1 point)** Is A diagonalizable? Justify your answer.

Solution: Yes, because there exists a basis of $\mathbb{R}^3$ consisting of the eigenvectors of A.

20. **(4 points)** Is the following matrix invertible? If it is invertible then find its inverse.

$$\begin{bmatrix} \frac{1}{2} & 2 & 0 \\ \frac{1}{2} & 3 & 1 \\ \frac{3}{2} & 6 & -3 \end{bmatrix}$$

Solution:

$$\left[\begin{array}{ccc|ccc} \frac{1}{2} & 2 & 0 & 1 & 0 & 0 \\ \frac{1}{2} & 3 & 1 & 0 & 1 & 0 \\ \frac{3}{2} & 6 & -3 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} \frac{1}{2} & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & -3 & -3 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{3}R_3} \left[\begin{array}{ccc|ccc} \frac{1}{2} & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -\frac{1}{3} \end{array}\right] \xrightarrow{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|ccc} \frac{1}{2} & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & \frac{1}{3} \\ 0 & 0 & 1 & 1 & 0 & -\frac{1}{3} \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|ccc} \frac{1}{2} & 0 & 0 & 5 & -2 & -\frac{2}{3} \\ 0 & 1 & 0 & -2 & 1 & \frac{1}{3} \\ 0 & 0 & 1 & 1 & 0 & -\frac{1}{3} \end{array}\right] \xrightarrow{R_1 \to 2R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 10 & -4 & -\frac{4}{3} \\ 0 & 1 & 0 & -2 & 1 & \frac{1}{3} \\ 0 & 0 & 1 & 1 & 0 & -\frac{1}{3} \end{array}\right]$$

The matrix is invertible and its inverse is $\begin{bmatrix} 10 & -4 & -\frac{4}{3} \\ -2 & 1 & \frac{1}{3} \\ 1 & 0 & -\frac{1}{3} \end{bmatrix}$

21. Let $A = \begin{bmatrix} 3 & 3 & 5 & -3 \\ 3 & 4 & 7 & -4 \\ -3 & -1 & -1 & 1 \end{bmatrix}$.

(a) **(3 points)** Find a basis for $\operatorname{Col} A$.

Solution:

$$\begin{bmatrix} 3 & 3 & 5 & -3 \\ 3 & 4 & 7 & -4 \\ -3 & -1 & -1 & 1 \end{bmatrix} \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 3 & 3 & 5 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 4 & -2 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 3 & 3 & 5 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 3R_2} \begin{bmatrix} 3 & 0 & -1 & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \to \frac{1}{3}R_1} \begin{bmatrix} 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Therefore a basis for $\operatorname{Col} A$ is given by $\left\{ \begin{bmatrix} 3 \\ 3 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix} \right\}$.

(b) **(3 points)** Find a basis for $\operatorname{Nul} A$.

Solution: from the R.E.F. of part (a) we have:

$$\begin{cases} x_1 = \frac{1}{3}x_3 \\ x_2 = -2x_3 + x_4 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{3}x_3 \\ -2x_3 + x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} \frac{1}{3} \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Therefore a basis for $\operatorname{Nul} A$ is $\left\{ \begin{bmatrix} \frac{1}{3} \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

22. An economy has two sectors: Government and Private. We know that:

- In order to produce one unit of output, the Government sector should consume $\frac{2}{3}$ units from itself and $\frac{1}{6}$ units from the Private sector.
- In order to produce one unit of output, the Private sector should consume $\frac{1}{5}$ units from itself and $\frac{2}{5}$ units from the Government sector.

(a) **(1 point)** Write down the consumption matrix for this economy.

Solution:
$C = \begin{bmatrix} \frac{2}{3} & \frac{2}{5} \\ \frac{1}{6} & \frac{1}{5} \end{bmatrix}$

(b) **(3 points)** Calculate the production levels required to meet a final demand of 6 units from the Government sector and 12 units from the Private sector.

Solution: $I - C = \begin{bmatrix} \frac{1}{3} & -\frac{2}{5} \\ -\frac{1}{6} & \frac{4}{5} \end{bmatrix} \Rightarrow (I-C)^{-1} = \frac{1}{\frac{3}{15}} \begin{bmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{1}{6} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ \frac{5}{6} & \frac{5}{3} \end{bmatrix}$

$\mathbf{x} = (I-C)^{-1}\mathbf{d} = \begin{bmatrix} 4 & 2 \\ \frac{5}{6} & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 6 \\ 12 \end{bmatrix} = \begin{bmatrix} 48 \\ 25 \end{bmatrix}$.